

14.7 CHANGE OF VARIABLES IN MULTIPLE INTEGRALS; JACOBIANS

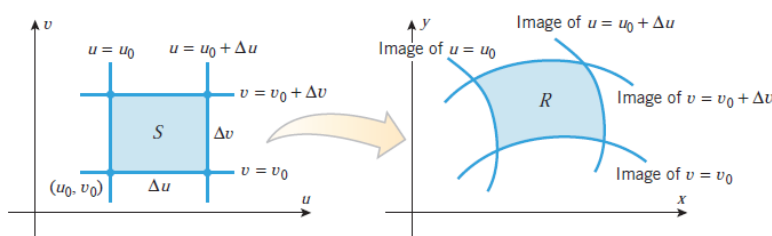
JACOBIANS IN TWO VARIABLES

To derive the change of variables formula for double integrals, we will need to understand the relationship between the area of a *small* rectangular region in the uv -plane and the area of its image in the xy -plane under a transformation T given by the equations

$$x = x(u, v), \quad y = y(u, v)$$

For this purpose, suppose that Δu and Δv are positive, and consider a rectangular region S in the uv -plane enclosed by the lines

$$u = u_0, \quad u = u_0 + \Delta u, \quad v = v_0, \quad v = v_0 + \Delta v$$



► Figure 14.7.5

14.7.1 DEFINITION If T is the transformation from the uv -plane to the xy -plane defined by the equations $x = x(u, v)$, $y = y(u, v)$, then the **Jacobian of T** is denoted by $J(u, v)$ or by $\partial(x, y)/\partial(u, v)$ and is defined by

$$J(u, v) = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v}$$

CHANGE OF VARIABLES IN DOUBLE INTEGRALS

Our next objective is to provide a geometric motivation for the following result.

14.7.2 CHANGE OF VARIABLES FORMULA FOR DOUBLE INTEGRALS If the transformation $x = x(u, v)$, $y = y(u, v)$ maps the region S in the uv -plane into the region R in the xy -plane, and if the Jacobian $\partial(x, y)/\partial(u, v)$ is nonzero and does not change sign on S , then with appropriate restrictions on the transformation and the regions it follows that

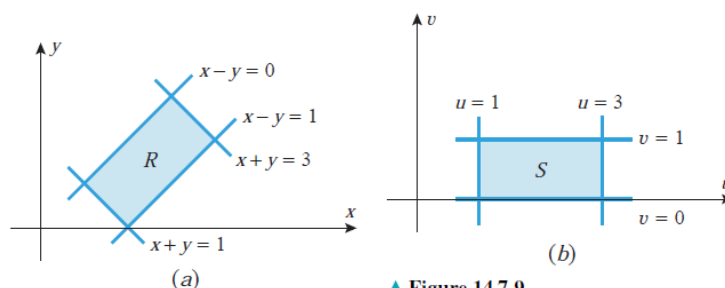
$$\iint_R f(x, y) dA_{xy} = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| dA_{uv} \quad (9)$$

where we have attached subscripts to the dA 's to help identify the associated variables.

► **Example 2** Evaluate

$$\iint_R \frac{x-y}{x+y} dA$$

where R is the region enclosed by $x - y = 0$, $x - y = 1$, $x + y = 1$, and $x + y = 3$ (Figure 14.7.9a).



▲ Figure 14.7.9

Solution. This integral would be tedious to evaluate directly because the region R is oriented in such a way that we would have to subdivide it and integrate over each part separately. However, the occurrence of the expressions $x - y$ and $x + y$ in the equations of the boundary suggests that the transformation

$$u = x + y, \quad v = x - y \quad (10)$$

would be helpful, since with this transformation the boundary lines

$$x + y = 1, \quad x + y = 3, \quad x - y = 0, \quad x - y = 1$$

are constant u -curves and constant v -curves corresponding to the lines

$$u = 1, \quad u = 3, \quad v = 0, \quad v = 1$$

in the uv -plane. These lines enclose the rectangular region S shown in Figure 14.7.9b. To find the Jacobian $\partial(x, y)/\partial(u, v)$ of this transformation, we first solve (10) for x and y in terms of u and v . This yields

$$x = \frac{1}{2}(u + v), \quad y = \frac{1}{2}(u - v)$$

from which we obtain

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$$

Thus, from Formula (9), but with the notation dA rather than dA_{xy} ,

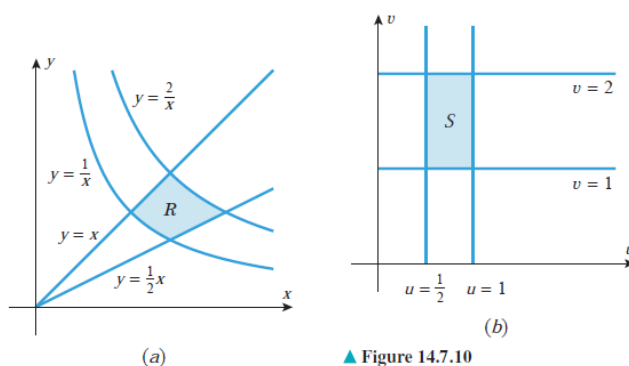
$$\begin{aligned}
 \iint_R \frac{x-y}{x+y} dA &= \iint_S \frac{v}{u} \left| \frac{\partial(x, y)}{\partial(u, v)} \right| dA_{uv} \\
 &= \iint_S \frac{v}{u} \left| -\frac{1}{2} \right| dA_{uv} = \frac{1}{2} \int_0^1 \int_1^3 \frac{v}{u} du dv \\
 &= \frac{1}{2} \int_0^1 v \ln |u| \Big|_{u=1}^3 dv \\
 &= \frac{1}{2} \ln 3 \int_0^1 v dv = \frac{1}{4} \ln 3 \quad \blacktriangleleft
 \end{aligned}$$

The underlying idea illustrated in Example 2 is to find a one-to-one transformation that maps a rectangle S in the uv -plane into the region R of integration, and then use that transformation as a substitution in the integral to produce an equivalent integral over S .

► **Example 3** Evaluate

$$\iint_R e^{xy} dA$$

where R is the region enclosed by the lines $y = \frac{1}{2}x$ and $y = x$ and the hyperbolas $y = 1/x$ and $y = 2/x$ (Figure 14.7.10a).



▲ Figure 14.7.10

Solution. As in the last example, we look for a transformation in which the boundary curves in the xy -plane become constant v -curves and constant u -curves. For this purpose we rewrite the four boundary curves as

$$\frac{y}{x} = \frac{1}{2}, \quad \frac{y}{x} = 1, \quad xy = 1, \quad xy = 2$$

which suggests the transformation

$$u = \frac{y}{x}, \quad v = xy \quad (11)$$

With this transformation the boundary curves in the xy -plane are constant u -curves and constant v -curves corresponding to the lines

$$u = \frac{1}{2}, \quad u = 1, \quad v = 1, \quad v = 2$$

in the uv -plane. These lines enclose the region S shown in Figure 14.7.10b. To find the Jacobian $\partial(x, y)/\partial(u, v)$ of this transformation, we first solve (11) for x and y in terms of u and v . This yields

$$x = \sqrt{v/u}, \quad y = \sqrt{uv}$$

from which we obtain

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} -\frac{1}{2u}\sqrt{\frac{v}{u}} & \frac{1}{2\sqrt{uv}} \\ \frac{1}{2}\sqrt{\frac{v}{u}} & \frac{1}{2}\sqrt{\frac{u}{v}} \end{vmatrix} = -\frac{1}{4u} - \frac{1}{4u} = -\frac{1}{2u}$$

Thus, from Formula (9), but with the notation dA rather than dA_{xy} ,

$$\begin{aligned} \iint_R e^{xy} dA &= \iint_S e^v \left| -\frac{1}{2u} \right| dA_{uv} = \frac{1}{2} \iint_S \frac{1}{u} e^v dA_{uv} \\ &= \frac{1}{2} \int_1^2 \int_{1/2}^1 \frac{1}{u} e^v du dv = \frac{1}{2} \int_1^2 e^v \ln |u| \Big|_{u=1/2}^1 dv \\ &= \frac{1}{2} \ln 2 \int_1^2 e^v dv = \frac{1}{2} (e^2 - e) \ln 2 \quad \blacktriangleleft \end{aligned}$$

■ CHANGE OF VARIABLES IN TRIPLE INTEGRALS

Equations of the form

$$x = x(u, v, w), \quad y = y(u, v, w), \quad z = z(u, v, w) \quad (12)$$

define a **transformation** T from uvw -space to xyz -space. Just as a transformation $x = x(u, v)$, $y = y(u, v)$ in two variables maps small rectangles in the uv -plane into curvilinear parallelograms in the xy -plane, so (12) maps small rectangular parallelepipeds

in uvw -space into curvilinear parallelepipeds in xyz -space (Figure 14.7.11). The definition of the Jacobian of (12) is similar to Definition 14.7.1.

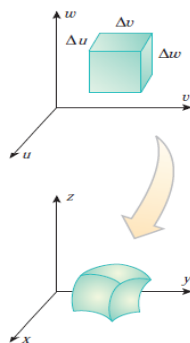


Figure 14.7.11

14.7.3 DEFINITION If T is the transformation from uvw -space to xyz -space defined by the equations $x = x(u, v, w)$, $y = y(u, v, w)$, $z = z(u, v, w)$, then the **Jacobian of T** is denoted by $J(u, v, w)$ or $\partial(x, y, z)/\partial(u, v, w)$ and is defined by

$$J(u, v, w) = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

For small values of Δu , Δv , and Δw , the volume ΔV of the curvilinear parallelepiped in Figure 14.7.11 is related to the volume $\Delta u \Delta v \Delta w$ of the rectangular parallelepiped by

$$\Delta V \approx \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| \Delta u \Delta v \Delta w \quad (13)$$

which is the analog of Formula (8). Using this relationship and an argument similar to the one that led to Formula (9), we can obtain the following result.

14.7.4 CHANGE OF VARIABLES FORMULA FOR TRIPLE INTEGRALS If the transformation $x = x(u, v, w)$, $y = y(u, v, w)$, $z = z(u, v, w)$ maps the region S in uvw -space into the region R in xyz -space, and if the Jacobian $\partial(x, y, z)/\partial(u, v, w)$ is nonzero and does not change sign on S , then with appropriate restrictions on the transformation and the regions it follows that

$$\iiint_R f(x, y, z) dV_{xyz} = \iiint_S f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| dV_{uvw} \quad (14)$$

► **Example 4** Find the volume of the region G enclosed by the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Solution. The volume V is given by the triple integral

$$V = \iiint_G dV$$

To evaluate this integral, we make the change of variables

$$x = au, \quad y = bv, \quad z = cw \quad (15)$$

which maps the region S in uvw -space enclosed by a sphere of radius 1 into the region G in xyz -space. This can be seen from (15) by noting that

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \text{becomes} \quad u^2 + v^2 + w^2 = 1$$

The Jacobian of (15) is

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = abc$$

Thus, from Formula (14), but with the notation dV rather than dV_{xyz} ,

$$V = \iiint_G dV = \iiint_S \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| dV_{uvw} = abc \iiint_S dV_{uvw}$$

The last integral is the volume enclosed by a sphere of radius 1, which we know to be $\frac{4}{3}\pi$. Thus, the volume enclosed by the ellipsoid is $V = \frac{4}{3}\pi abc$. ◀

Note: Jacobians also arise in converting triple integrals in rectangular coordinates to iterated integrals in cylindrical and spherical coordinates.

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

is

$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r$$

and the Jacobian of the transformation

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

is

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \rho^2 \sin \phi$$

Such that

$$\iiint_G f(x, y, z) dV = \iiint_{\text{appropriate limits}} f(r \cos \theta, r \sin \theta, z) \frac{\partial(x, y, z)}{\partial(r, \theta, z)} dz dr d\theta \quad (16)$$

$$\iiint_G f(x, y, z) dV = \iiint_{\text{appropriate limits}} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} d\rho d\phi d\theta \quad (17)$$

The absolute-value signs are omitted from Formulas (16) and (17) because the Jacobians are nonnegative.

Practice Problem:

Chapter-14.7:

Problems: 3,7,11,16,21-30,35,36

Thank you for your attention!